Seminar 4 - Quiz.

① **A.** $\cos \sphericalangle(a,b) = \dfrac{\langle a,b\rangle}{|a|\cdot|b|}$

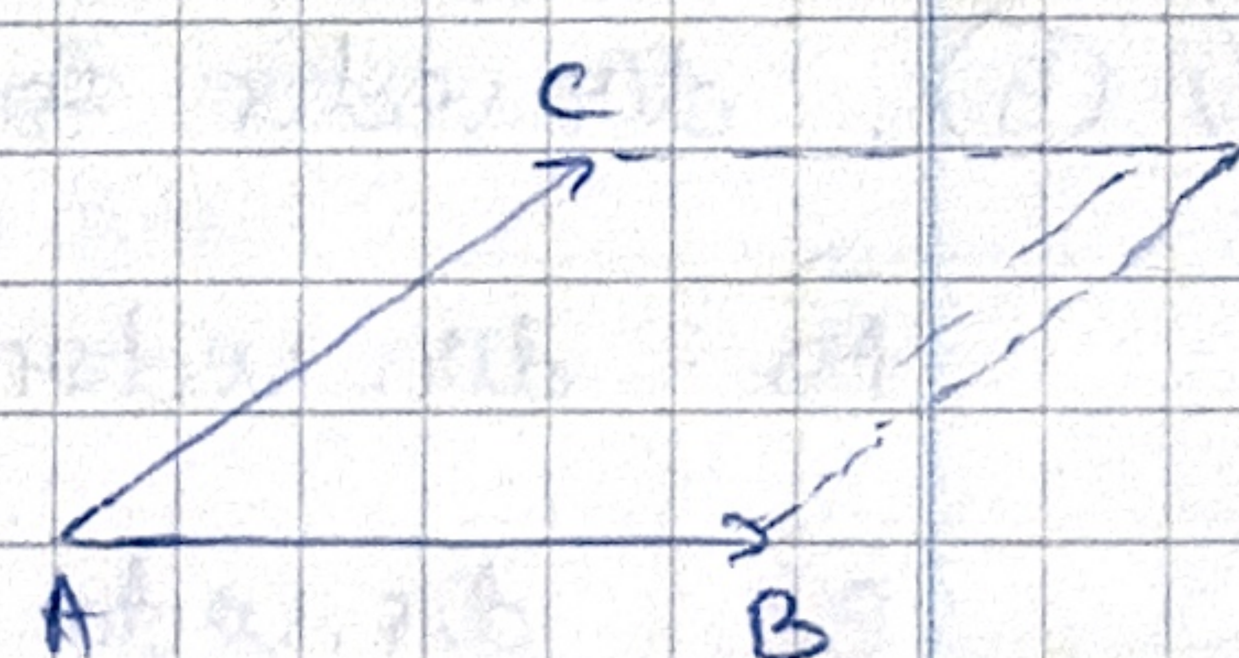
B. $\sin \sphericalangle_{or}(a,b) = \dfrac{[a,b]}{|a|\cdot|b|}$

C. $\tan \sphericalangle_{or}(a,b) = \dfrac{[a,b]}{\cancel{|a|\cdot|b|}} \cdot \dfrac{\cancel{|a|\cdot|b|}}{\langle a,b\rangle} = \dfrac{[a,b]}{\langle a,b\rangle}$

D. $\cot \sphericalangle_{or}(a,b) = \dfrac{\langle a,b\rangle}{[a,b]} = \dfrac{\langle a,b\rangle}{[a,b]}$

②. $\vec{AB}\,(6,0,1)$

$\vec{AC}\,(1{,}5, 2, 1)$



$\text{Area}(ABCD) = |AB \times AC|$.

$$AB \times AC = \begin{vmatrix} i & j & k \\ 6 & 0 & 1 \\ 1{,}5 & 2 & 1 \end{vmatrix} = \ldots = (-2, -4{,}5, 12)$$

$$\text{Area}(ABCD) = |(-2, -4{,}5, 12)| = \sqrt{2^2 + (4{,}5)^2 + 12^2} = 12{,}971.$$

$$\text{Area}(ABCD) = b \cdot h \Rightarrow h = \frac{\text{Area}}{b} = \frac{12{,}971}{|AB|} = \frac{12{,}971}{\sqrt{37}} = 2{,}132.$$

$$|AB| = \sqrt{6^2 + 1^2} = \sqrt{37}$$

Setup1: A, B, C, D. → K.

③. $A_{\triangle ABC} = \dfrac{1}{2}|AB \times AC| = \dfrac{1}{2}\sqrt{4^2 + (-2)^2} = \dfrac{1}{2} \cdot 2\sqrt{5} = \sqrt{5}.$

$$\vec{AB} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \quad \vec{AC} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}; \quad AB \times AC = \begin{vmatrix} i & j & k \\ 0 & 2 & 0 \\ 1 & 3 & -2 \end{vmatrix} = (4, 0, -2)$$

✓ (4) $V_{ABCD} = \frac{1}{6} \left\| \begin{matrix} \vec{AB} \\ \vec{AC} \\ \vec{AD} \end{matrix} \right\| = \frac{1}{6} \cdot \cancel{14}\,6 \cdot \cancel{\frac{7}{3}} \cdot \frac{|-6|}{6} = \frac{6}{6} = 1$

$\begin{vmatrix} 0 & 2 & 0 \\ 1 & 3 & -2 \\ 1 & 2 & 1 \end{vmatrix} = 2\cdot(-1)^2 \cdot \begin{vmatrix} 3 & -2 \\ 2 & 1 \end{vmatrix} = \cancel{2\cdot(3+4) = 14}.\ -6.$

$\vec{AB} = (0, 2, 0)$

$\vec{AC} = (1, 3, -2)$

$\vec{AD} = (1, 2, 1)$

✓ (5). dir. vector for the common ⊥ of AD, BC.

$\vec{AD}$: dir. vector : $D - A = (1, 2, 1) = v_1$.

$\vec{BC}$: dir. vector : $C - B = (1, 1, -2) = v_2$.

dir. of common ⊥ $m = v_1 \times v_2$.

$m = v_1 \times v_2 = \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 1 & 1 & -2 \end{vmatrix} = (-5, 3, -1)$.

✓ (6). $d(AB, CD)$

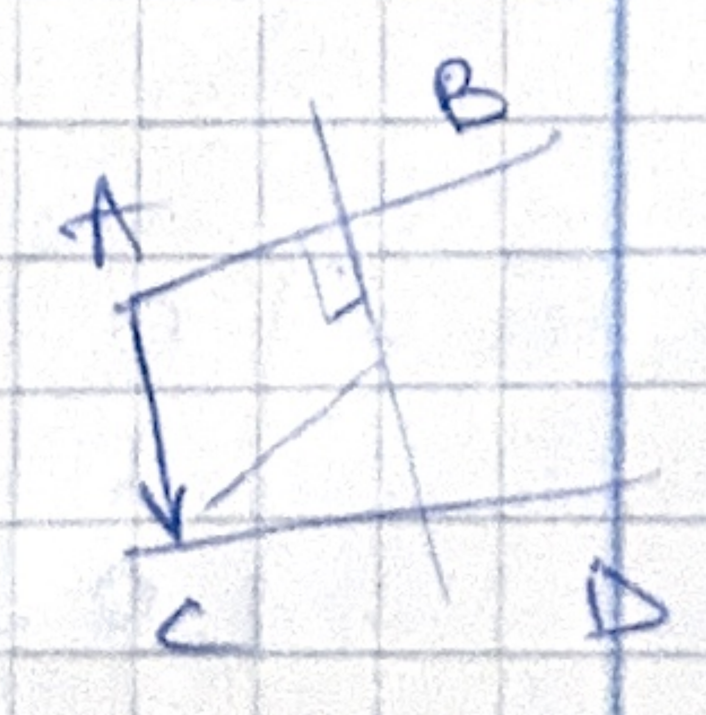
$u = \vec{AB} = (0, 2, 0)$; $A(-1, -1, 1)$.

$v = \vec{CD} = (0, 1, -3)$, $C(0, 2, -1)$.

dir. vectors $(0,2,0)$, $(0,1,-3)$ not proportional

$\Rightarrow AB \nparallel CD. \Rightarrow$ skew.

$u \times v = \begin{vmatrix} i & j & k \\ 0 & 2 & 0 \\ 0 & 1 & -3 \end{vmatrix} = (6, 0, 0)$.



$$d = \frac{|\langle AC, u \times v \rangle|}{|u \times v|} = \frac{\sqrt{6^2}}{\sqrt{6^2}} = \frac{6}{6} = 1.$$

$AC = (1,3,-2)$.

$\langle AC, u\times v\rangle = \langle (1,3,-2), (6,0,0)\rangle =$

$= 6$

! ✓ 7. Cartesian eq. for the common $\perp$ line of AB, CD.

$\vec{AB}$: dir. vector: $(0,2,0) = u$

$\vec{CD}$: dir. vector: $(0,-1,3) = v$

dir. of common $\perp$:

$m = u \times v = (1,2,1) \times (0,-1,3) = (6,0,0)$

$$= \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 0 & -1 & 3 \end{vmatrix} = i \cdot \begin{vmatrix} 2 & 1 \\ -1 & 3 \end{vmatrix} - j \cdot \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} + k \cdot \begin{vmatrix} 1 & 2 \\ 0 & -1 \end{vmatrix} =$$

$= i \cdot (6+1) - j \cdot 3 + k(-1) = 7i - 3j - k.$

$= (7,-3,-1).$

$\cancel{7x - 3y - 1 = 0}$

$\sim m = (1,0,0)$

π_1: contains line AB and the $\perp$ dir. m.

π_2: " — " — CD " — " —

- $A(-1,-1,1)$.

$$m_1 = u \times m = (0,2,0) \times (1,0,0)$$

$$m_1 = \begin{vmatrix} i & j & k \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{vmatrix} = (0,0,-2).$$

$$\Rightarrow \quad 0\cdot(x - x_A) + 0\cdot(y - y_A) + (-2)\cdot(z - z_A) = 0$$

$$\Rightarrow \quad 0\cdot(x+1) + 0\cdot(y+1) - 2(z-1) = 0$$

$$\Rightarrow \quad -2z + 2 = 0 \Rightarrow \underline{z = 1}.$$

- $C(0,2,-1)$.

$$m_2 = v \times m = (0,-1,3) \times (1,0,0)$$

$$m_2 = \begin{vmatrix} i & j & k \\ 0 & -1 & 3 \\ 1 & 0 & 0 \end{vmatrix} = (0,3,1).$$

$$\Rightarrow \quad 0(x-0) + 3\cdot(y-2) + 1\cdot(z+1) = 0$$

$$\Rightarrow \quad 3y - 6 + z + 1 = 0$$

$$\Rightarrow \quad 3y + z - 5 = 0.$$

$$\begin{cases} \cancel{z=1} \quad \pi_1 : z - 1 = 0 \\ \pi_2 : 3y + z - 5 = 0 \end{cases} \Rightarrow \begin{cases} 3y - 4 = 0. \\ z - 1 = 0. \end{cases}$$

8. $d(D, (ABC))$.

$\pi: ax + by + cz + d = 0.$

$\vec{AB} = (0,2,0)$

$\vec{AC} = (1,3,-2).$

$$n = \vec{AB} \times \vec{AC} = \begin{vmatrix} i & j & k \\ 0 & 2 & 0 \\ 1 & 3 & -2 \end{vmatrix} = (-4,0,-2) \sim (2,0,1).$$

- $A(-1,-1,1)$, $n(2,0,1)$.

$(ABC): 2(x+1) + 0\cdot(y+1) + 1\cdot(z-1) = 0.$

$\Rightarrow 2x + z + 1 = 0.$

$$d(D,(ABC)) = \frac{|a\cdot x_D + b\cdot y_D + c\cdot z_D + d|}{\sqrt{a^2+b^2+c^2}} =$$

$$= \frac{|2\cdot 0 + 0\cdot 1 + 1\cdot 0 + 1|}{\sqrt{2^2+0^2+1^2}} = \frac{|3|}{\sqrt{5}} = \frac{3}{\sqrt{5}}.$$

9. vector product $\boxed{A}$ billinear

$\boxed{B}$ skew-sym.

10. $V_{ABCD} = \frac{1}{6}\left\|\begin{matrix}\vec{AB}\\ \vec{AC}\\ \vec{AD}\end{matrix}\right\| = 5.$

$\vec{AB} = B - A = (1,-1,2)$

$\vec{AC} = (0,-2,4)$

$\vec{AD} = (x_D-2, y_D-1, z_D+1).$

$V = \frac{1}{6} \left| -z - 2y + 1 \right| = 5$

$\Rightarrow \quad |2 - 4y| = 30$

$$\begin{vmatrix} 1 & -1 & 2 \\ 0 & -2 & 4 \\ x-2 & y-1 & z+1 \end{vmatrix} \sim \begin{vmatrix} 1 & -1 & 2 \\ 0 & -1 & 2 \\ x-2 & y-1 & z+1 \end{vmatrix} \sim \begin{vmatrix} 1 & 0 & 0 \\ 0 & -1 & 2 \\ x-2 & y-1 & z+1 \end{vmatrix}$$

$$\sim 1 \cdot (-1)^2 \cdot \begin{vmatrix} -1 & 2 \\ y-1 & z+1 \end{vmatrix} = -z - 1 - 2(y-1) =$$

$$= -z - 2y - 1 + 2 = -z - 2y + 1. = -2y + 1.$$

$\sim 2 - 4y.$

$D \in Oy \Rightarrow (0, y, 0).$

a: $2 - 4y = 30 \Rightarrow 4y = -28 \Rightarrow y = -7.$

ou: $2 - 4y = -30 \Rightarrow y = 8.$

$\Rightarrow D(0, -7, 0)$ or $D(0, 8, 0)$.